

# AI Engineer Career Roadmap

|                        |                                    |                         |            |
|------------------------|------------------------------------|-------------------------|------------|
| ■ Working Professional | ■ Python · Web Dev · Data Analysis | ■ Become an AI Engineer | ■ 3 Months |
|------------------------|------------------------------------|-------------------------|------------|

■ CURRENT POSITION

■ SKILL GAP ANALYSIS

|   |               |                      | Skill                       | Status  | Priority |
|---|---------------|----------------------|-----------------------------|---------|----------|
| ■ | Role          | Working Professional | ML / Deep Learning          | Missing | ■ High   |
|   | Python        | Intermediate         | LLM APIs (OpenAI/Anthropic) | Missing | ■ High   |
|   | Web Dev       | Intermediate         | Vector DBs / RAG            | Missing | ■ Med    |
|   | Data Analysis | Intermediate         | MLOps / Deployment          | Missing | ■ Med    |
|   |               |                      | Prompt Engineering          | Missing | ■ Low    |

■ RECOMMENDED LEARNING PLAN

| Phase            | Focus Area                           | Resources                       | Duration    |
|------------------|--------------------------------------|---------------------------------|-------------|
| 1 — Foundation   | ML Basics, Scikit-learn, NumPy       | fast.ai, Kaggle Learn           | Weeks 1–3   |
| 2 — LLMs & APIs  | OpenAI API, LangChain, Prompt Eng.   | DeepLearning.AI short courses   | Weeks 4–6   |
| 3 — RAG & Agents | Pinecone / ChromaDB, AI Agents       | LlamaIndex docs, YouTube        | Weeks 7–9   |
| 4 — Deployment   | FastAPI, Docker, Hugging Face Spaces | Official docs + GitHub projects | Weeks 10–12 |

■ SUGGESTED PROJECTS

■ NETWORKING STRATEGY

|   |               |                                                           |            |                                            |                                                           |
|---|---------------|-----------------------------------------------------------|------------|--------------------------------------------|-----------------------------------------------------------|
| ■ | AI Chatbot    | LangChain + OpenAI + Streamlit — deploy on HF Spaces      | ■ LinkedIn | Post weekly AI project updates + learnings |                                                           |
| ■ | RAG Pipeline  | Ingest PDFs, embed with ChromaDB, query with LLM          |            | ■ GitHub                                   | Push every project — green squares = credibility          |
| ■ | Data + AI App | Data analysis + GPT-4 insights dashboard (your strength!) |            | ■ Communities                              | Join Hugging Face Discord, AI Twitter/X, local meetups    |
| ■ | REST AI API   | FastAPI wrapper around an LLM with auth + rate limiting   |            | ■ Target DMs                               | Connect with AI Engineers at startups — offer value first |

■ MONTHLY MILESTONES

| Month 1                                                            | Month 2                                                              | Month 3                                                                    |
|--------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| ■ Complete ML basics ■ Call 3 LLM APIs ■ Ship chatbot v1 on GitHub | ■ Build RAG pipeline ■ Launch on HF Spaces ■ Post 8 LinkedIn updates | ■ Deploy FastAPI AI app ■ Apply to 10+ AI roles ■ 3 warm referrals secured |

■ IMMEDIATE NEXT ACTIONS (This Week)

|         |                                                                     |
|---------|---------------------------------------------------------------------|
| ■ Today | Sign up for fast.ai course & OpenAI API key                         |
| ■ Day 2 | Create a GitHub repo named 'ai-engineer-journey'                    |
| ■ Day 3 | Run your first LLM API call using Python                            |
| ■ Day 5 | Update LinkedIn headline → 'Python Dev   Aspiring AI Engineer'      |
| ■ Day 7 | Publish LinkedIn post: 'I'm learning AI in 3 months — follow along' |